EE2004 Digital Signal Processing

Tutorial 8 Answer Key

1. Hint: Simplify $1 - A(z) \cdot A^*(z)$
2. Inequality reverses
3. $0 < |a| < \frac{1}{6}$, $a \in \mathbb{R}$
4. Sequence (h) has all its roots inside the unit circle
5. Sequence (h) has the most rapid increase of its partial energy.
   Sequence (c) has the least rapid increase of its partial energy